

Kyung-Min Jin

CONTACT INFORMATION

LG Electornics
Multimodal Task
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SUMMARY

Building multimodal intelligence from visual perception to embodied reasoning.

My research began in the computer vision domain, where I designed novel pose estimation frameworks that combine transformer-based architectures with convolutional neural networks. This work achieved state-of-the-art performance across multiple benchmarks and led to publications in international conferences and peer-reviewed journals, including WACV.

After joining LG Electronics, I worked on deploying body and hand pose estimation models to edge devices, focusing on efficiency and robustness in real-world environments. As my research interests expanded, I transitioned toward multimodal learning, contributing to the development of large-scale Vision–Language–Audio models, cross-modal continual learning strategies, and reinforcement learning–based policy optimization methods such as GRPO and DPO.

More recently, I have been working within an advanced robotics research team, where my focus is on robotic perception and natural language–based object understanding. My current research aims to bridge multimodal foundation models with embodied AI, enabling robots to ground language in visual perception and interact with the physical world more effectively.

INDUSTRIAL EXPERIENCE

LG Electronics

Seoul, Korea

Advanced Robotics Lab. Robot Cognition Solution Task Specialist

2026.01 - Present

Developed of Natural Language-Based Object Understanding for Robots

- Construction of a contextual Knowledge Graph and RAG-based Vector Database
- Generated of QA pairs based on shots & history
- Developed of a time-sensitive hybrid RAG QA system
- *Skills - Python, Pytorch, LangChain, llama-index, Qdrant*

AI Lab. Multimodal Task Specialist

Jul. 2023.07 - Dec. 2025

Researched "Semantic-Aware Mutual Information Factorized Learning (SAMIF)"

- Designed, experimented, and validated cross-modality shared, unique information factorized learning
- Achieved state-of-the-art performance for multimodal semantic segmentation (2 – 3% ↑)
- *Skills - Python, Pytorch, Cross-attention*

Distributed semantic chunking and contextual trigger

- Collected, refined, and analyzed an aligned home appliance log, image, audio, and video data
- Developed and distributed a time-series forecasting model that analyzed and identified patterns using SAMIF
- Transferred and distributed an on-device real-time action detection and appliance control model to Pytorch → onnx → Qualcomm Neural Processing SDK
- Made PoC by improving the IoU (0.3 ↑) and f1-score (23% ↑) in action localization
- *Skills - Python, Pytorch, Cross-attention, ONNX, Qualcomm SDK*

Generalized a personalized home automation service

- Created VectorDB with chunked video summaries, images, and home appliance logs
- Designed and developed multimodal RAG system and demo web-app
- *Skills - Python, Pytorch, HuggingFace Transformers, Cross-attention, docker-compose, LangChain, llama-index, chromadb, gradio*

Trained and developed an small multimodal LLM

- Designed an audio tower and LLM cross modal continual learning pipeline
- Generated Multimodal Korean DPO dataset and researched spatial reasoning
- Achieved a **20.7% improvement** by training GRPO on synthetic data with Self-Reflection and Unification
- MLLM demo app serving within the company - image and video-based QA
- *Skills - Python, Pytorch, Cross-attention, docker-compose, FastAPI, Continual Learning, TRL, gradio*

Researched and Developed of Fast-Reasoning Modality-Aware Policy Optimization

- Proposed a loss update and reward mechanism that applies modality-aware online weighting in GRPO
- Improved answer accuracy after fast reasoning by predicting missed and hallucinated cases
- Collaborated with the Toronto AI Lab to conduct research and wrote a paper
- *Skills - Python, Pytorch, veRL*

Cognitive Vision Researcher

Mar. 2023 - Jul. 2023

Optimized body pose estimation

- Trained 2D pose estimation & 3D pose estimation using quantized-aware-training
- Introduced body pose estimation to edge device (LG TV)
- *Skills - Python, Pytorch, transformers, Quantized-aware-training, ONNX*

Researched hand pose estimation

- Designed and experimented RecycleNet, re-trains the synthesized hand mesh
- Proposed and experimented grouping attention of each finger
- Achieved state-of-the-art performance in PA-MPJPE, wrote a paper
- *Skills - Python, Pytorch, transformers*

Da Vinci Co., Ltd.

Seoul, Korea

Frontend developer

Jul. 2020 - Feb. 2021

Deployed and maintained map-based app, link saving app, and chrome extension

- *Skills - React, Typescript, Next.js, React Native*

Klue

Seoul, Korea

Frontend developer

Jul. 2019 - Mar. 2023

Designed, deployed, and maintained a course evaluation service

- *Skills - React, Typescript, Redux, MobX*

RESEARCH EXPERIENCE	Voice Caddie	Seoul, Korea
	AI Assistant Device Developer	May 2020 - Nov. 2020
	<u>Deployed golf pose estimation model, action localization, and annotation tool</u>	
	• Skills - Python, Pytorch, C++, OpenCV	
EDUCATION	Korea University	Seoul, Korea
	Department of Artificial Intelligence	Sep. 2021 - Feb. 2023
	• M.S. in Artificial Intelligence • Thesis: Kinematic Continuity-Aware Masked Hierarchical Attention for Pose Estimation in Videos • Adviser: Seong-Whan Lee • Laboratory: PRML Laboratory • Area of Study: Computer Vision • Total GPA: 4.50 / 4.50 (100 / 100)	
	College of Informatics	Mar. 2016 - Aug. 2021
	• B.S. in Computer Science & Engineering • B.S. in Artificial Intelligence • Total GPA: 3.99 / 4.50 (94.2 / 100)	GPA: 4.02 / 4.50 (94.5 / 100) GPA: 4.17 / 4.50 (96.2 / 100)
PUBLICATIONS	[1] Nikhil Verma*, Minjung Kim*, JooYoung Yoo*, Kyung-Min Jin*, Manasa Bharadwaj, Kevin Ferreira, Ko Keun Kim, and Youngjoon Kim, "MAPLE: Modality-Aware Post-training and Learning Ecosystem," in <i>arXiv preprint</i>, 2026. [2] Tae-Kyung Kang, Gun-Hee Lee, Woo-Jeoung Nam, Hyun-Woo Kim, Kyung-Min Jin, and Seong-Whan Lee, "Calibrated Attention Masking Network for Temporal Action Localization," Under Review. [3] Kyung-Min Jin, Gun-Hee Lee, Woo-Jeoung Nam, Tae-Kyung Kang, Hyun-Woo Kim, and Seong-Whan Lee, "Masked Kinematic Correlation with Hierarchical Attention for Pose Estimation," in <i>Neural Netw. (NN)</i>, 2024. [4] Hyun-Woo Kim, Gun-Hee Lee, Woo-Jeoung Nam, Kyung-Min Jin, Tae-Kyung Kang, Geon-Jun Yang, and Seong-Whan Lee, "MHCanonNet: "Multi-Hypothesis Canonical Lifting Network for Self-supervised 3D Human Pose Estimation in the wild Video," in <i>Pattern Recognit. (PR)</i>, 2024. [5] Bo-sang Kim, Jong-hyun Kim, Hyo-tae Lee, Lan-ying Jin, Jeong-won Ha, Do-woo Kwon, Jung-pyo Kim, Won-hyeok Im, Kyung-Min Jin, Jung-ho Lee, "Mesh Represented Recycle Learning for 3D Hand Pose and Mesh Estimation," in <i>arXiv preprint</i>, 2023. [6] Kyung-Min Jin, Byoung-Sung Lim, Gun-Hee Lee, Tae-Kyung Kang, and Seong-Whan Lee, "Kinematic-aware Hierarchical Attention Network for Human Pose Estimation in Videos," in <i>Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)</i>, 2023. (Oral) [7] Tae-Kyung Kang, Gun-Hee Lee, Kyung-Min Jin, and Seong-Whan Lee, "Action-aware Masking Network with Group-based Attention for Temporal Action Localization," in <i>Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)</i>, 2023. (Oral) [8] Kyung-Min Jin, Gun-Hee Lee, and Seong-Whan Lee, "OTPose: Occlusion-Aware Transformer for Pose Estimation in Sparsely-Labeled Videos," in <i>Proc. IEEE Int. Conf. Syst. Man Cybern. (SMC)</i>, 2022. (Oral)	
	* Equal contribution	

ACADEMIC SERVICE	Reviewer of NeurIPS (Conference on Neural Information Processing Systems)	2026
	Reviewer of BMVC (British Machine Vision Conference)	2026
	Reviewer of ECCV (European Conference on Computer Vision)	2026
	Reviewer of CVPR (IEEE Conference on Computer Vision and Pattern Recognition)	2024 - 2026
	Reviewer of ICCV (International Conference on Computer Vision)	2025
GRANTS AND HONORS	[1] Awarded R&D Scholarships - LG Electronics	Sep. 2022
	[2] Awarded Grand Prize - Miso AI Model Development Challenge - Ministry of Science and ICT Infant Behavior Video Data hosted by	Dec. 2021
	[3] Encouragement Prize Smart Campus Dataton 2020 - Korea University	Jul. 2020)
TEACHING EXPERIENCE	Teaching Assistant	Korea University
	• Teaching Assistant for COSE213-02: Data Structure	Sep. 2022 - Dec. 2022
	Instructor	Inflearn
	• Online 2D Pose Estimation Course	Mar. 2023
CERTIFICATE	Human Resources Development Service of Korea	Seoul, Korea
	• Engineer Information Processing	Nov. 2021
SKILLS	Computer Programming	
	• Python, JavaScript, and C	
	Deep Learning Frameworks	
	• PyTorch, Tensorflow, Keras, and Qualcomm Neural Processing SDK	